

Design and Implementation of a Residential Air Pollution Detector

Introduction

Indoor air pollution can be cause for numerous health concerns such as asthma, fatigue, and cognitive decline. Existing solutions to indoor air pollution in residential settings include HEPA filters, which are very effective at removing particulate matter but unable to remove gaseous pollutants. Another existing solution is activated carbon purifiers, which can remove VOCs and gases effectively, though they're not fully effective with carbon monoxide, ammonia, bacteria, arsenic, nitrates, and some other metals.

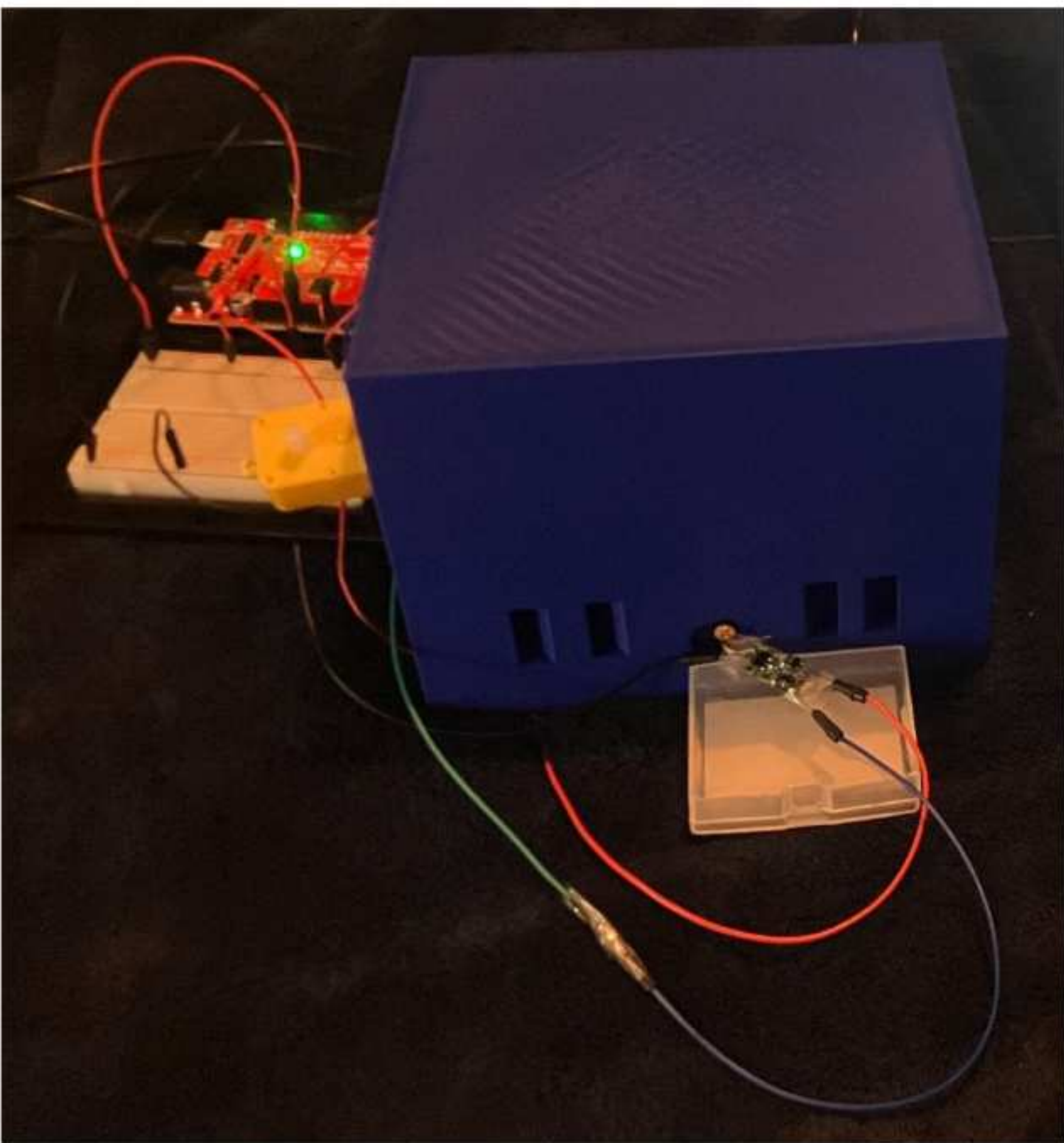
This project aims to create a reliable Ammonia detector with at least 80% accuracy. It should fall below \$250, as the detector should be affordable to be open to installment in public schools or homes. The project will be catered to residential settings, and so it will be cost effective, totaling below \$250 for the prototype construction. It involves shining a 1530 nm laser through air into a photodiode sensor, which sends readings through a transimpedance amplifier and arduino into a computer program, which alerts the user if Ammonia is highly detected. The laser and sensor are contained in a 3D printed box created through TinkerCAD, which includes a fan and tilted vents for air circulation. The computer algorithm is able to accurately read in sample data and determine if ammonia is present when readings drop below the determined threshold.

Methods

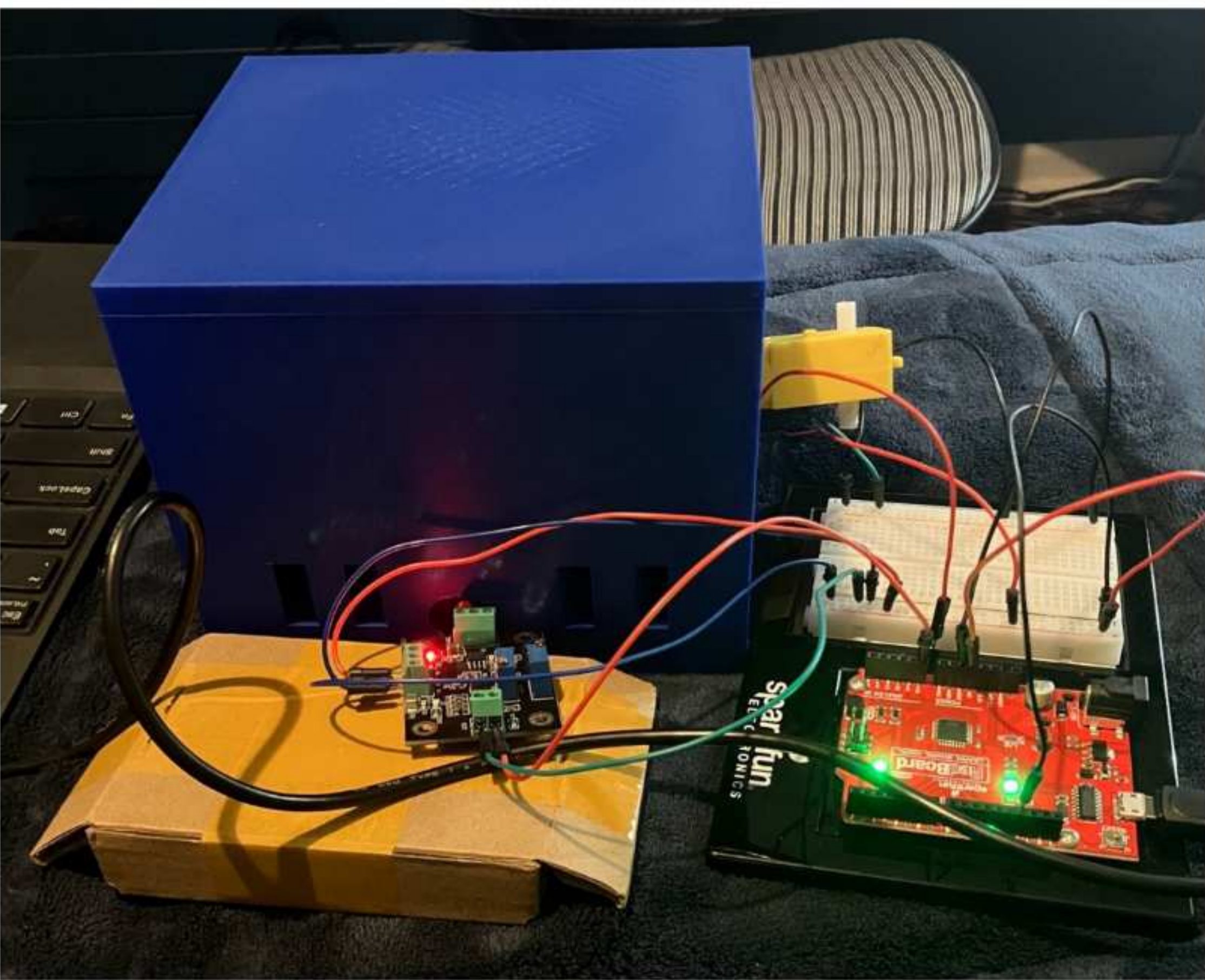
The detector will involve shining lasers through air into a photodiode sensor. A 1530 nm laser will be used initially because Ammonia's light spectrum exhibits weak overtone absorption lines around near 1530 nm in the NIR light spectrum. This will make detection clear because, when present, Ammonia will absorb some of the 1530 nm light, so less light will be sensed. The photodiode sensor will be a 1 mm Photosensitive Surface InGaAs PIN/PD Photodiode Detector, capable of sensing 1530 nm. Ammonia is the initial focus because it is a common pollutant that is neglected by existing HEPA and activated carbon filters.

The detector's casing is modelled in TinkerCAD and 3d-printed, including vents tilted away from the photodiode sensor to avoid external light impacting the sensor's readings. The casing will also have holes fitted to hold the fan, laser, and photodiode sensor in place. The fan will blow air around the filter to improve circulation. The photodiode sensor will be connected to a Current to Voltage Photodiode Amplifier Module, manufactured by Senzoee, to convert the signal from current into voltage. The amplifier will send data to an arduino, which will wire to log data in a Windows laptop arduino software program. Once the laptop is reading the data, it will run it through an Arduino IDE algorithm that will identify if Ammonia (or other pollutants) are present in the air. The algorithm will use a simple threshold calculation. This involves the program checking if the reading drops below a certain threshold, which would signify Ammonia. The threshold will be determined by observing the sensors readings before and after Ammonia is sprayed into the filter. The program takes in the readings and outputs the boolean value of if Ammonia or detected or not.

The filter's accuracy will be tested with common household sources of Ammonia, to simulate residential conditions. Cleaning products will be sprayed into the air and, if the filter is successful, it will identify Ammonia as present. The spray will be Windex Glass Cleaner, a popular cleaning spray that contains Ammonium hydroxide. In addition, a control spray of Windex Ammonia-free will be included to improve reliability, where the filter should not identify ammonia as present. After each spray the fan should eventually clear the filter of any sprays, which will prepare it for the next test. 10 tests should be done and logged to record accuracy.

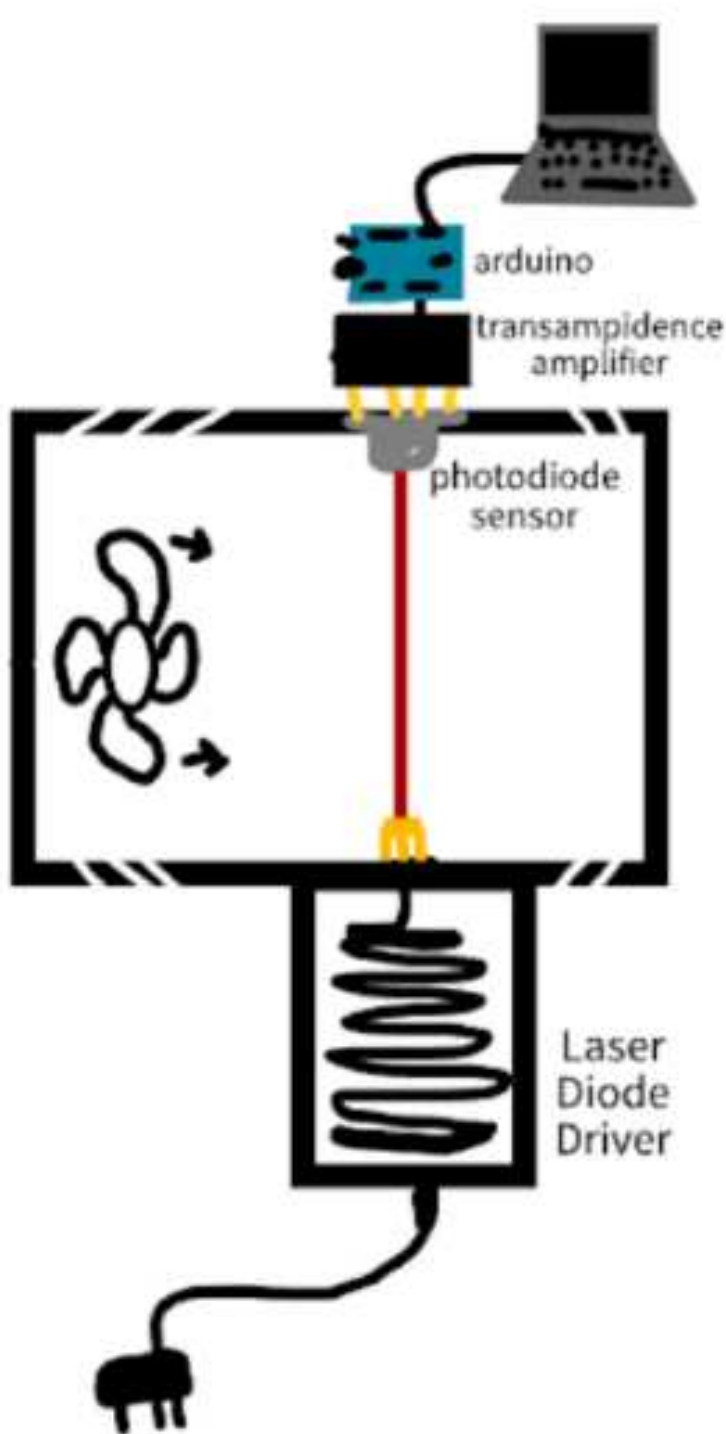


Laser Side

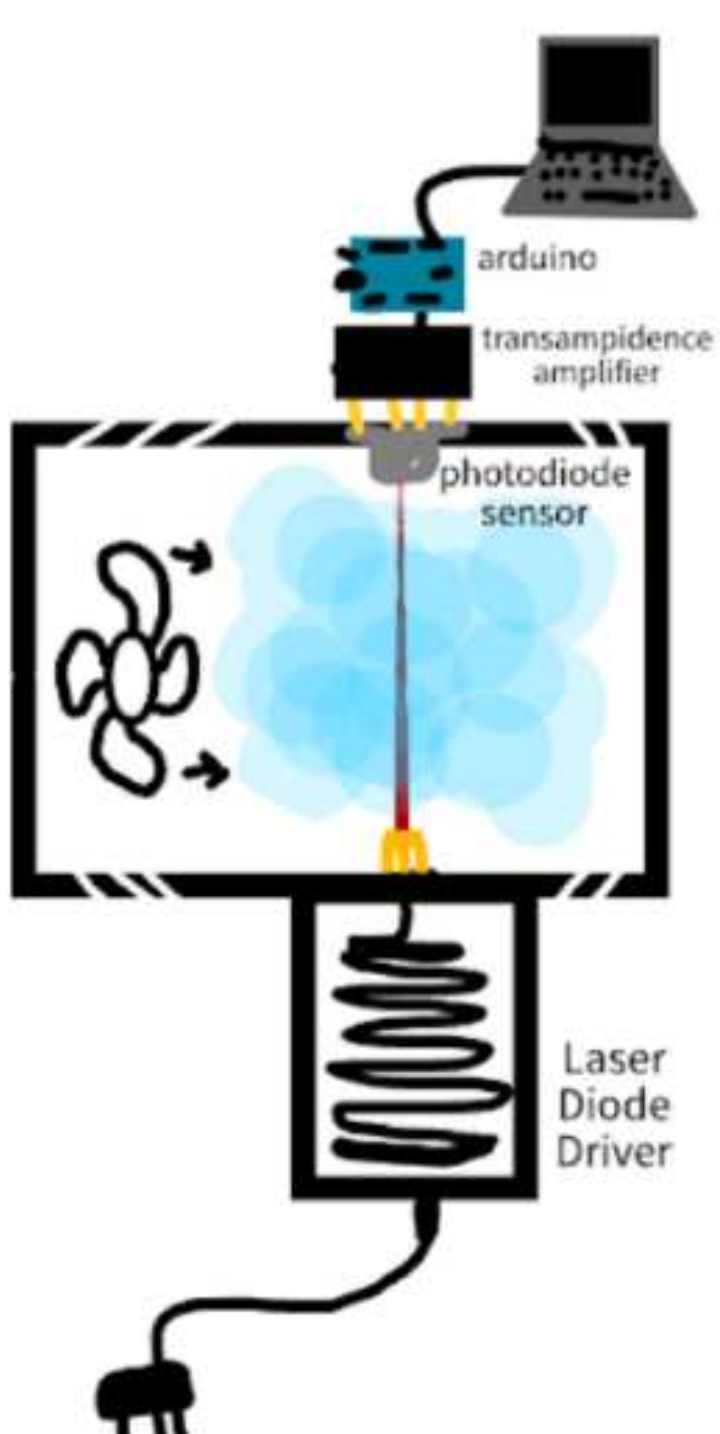


Photodiode Sensor Side

No ammonia



ammonia



Results

The overall design was able to come in under \$250, significantly below other Ammonia detection methods, such as a full infrared spectroscopy system. This makes the detector accessible in schools and homes with conservative budgets. The casing of the detector was successfully modelled to scale in TinkerCad, shown above. Upon testing the best threshold was found to be 135, where readings below that represent Ammonia being detected. The detector was tested with Windex Glass Cleaner sprayed through the vents for 10 trials, each separated by a minimum of 5 minutes to let the filter ventilate. Afterwards, the detector was tested with Windex Ammonia-free, again for 10 trials with a minimum period of 5 minutes between trials. The resulting data is shown below. During testing with Windex containing ammonia, ammonia was correctly identified as present in 7 out of 10 trials (70% accuracy). When ammonia-free Windex was tested, ammonia was correctly identified as not present in 6/10 trials.

Table 1. Windex Glass Cleaner Readings

Trial	Reading	Ammonia detected?
1	136	No
2	134	Yes
3	133	Yes
4	134	Yes
5	132	Yes
6	134	Yes
7	134	Yes
8	136	No
9	134	Yes
10	135	No

Table 2. Windex Ammonia-free Readings

Trial	Reading	Ammonia detected?
1	137	No
2	133	Yes
3	135	No
4	136	No
5	132	Yes
6	135	No
7	134	Yes
8	134	Yes
9	137	No
10	135	No

Discussion + Future Research

The overall design successfully under \$250, significantly below other Ammonia detection methods, such as a full infrared spectroscopy system. This makes the detector accessible in schools and homes with conservative budgets. The casing of the detector was successfully modelled to scale in TinkerCad, and 3D printed. The algorithm was determined to operate best at a threshold of 135, detecting ammonia as present when readings drop below 135. The data indicates the detector is somewhat capable in identifying ammonia as present when compared to clean air, but is less capable in differentiating when ammonia is in the air as opposed to other pollutants. It could be implemented in public schools as a rough estimate of the presence of Ammonia, yet accuracy should improve for it to be considered reliable in its detection.

Future research would involve testing more complex Ammonia identification software algorithms. These would consider patterns in readings over time. For example, if the filter was moved from one place to another the base lighting readings would change. This would require the algorithm to adjust the threshold based on what the baseline is in the environment of the filter. Other factors, such as lights outside of the filter turning off during nighttime, would be accounted for to improve the filter. Future research would also look at expanding the scope of the filter to include other common gas pollutants, such as Nitrogen Monoxide. The detector would expand by using additional lasers, correlated with the gas that absorbs the laser wavelength.

References

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Limitations

The detection accuracy was variable due to fluctuations in airflow within the chamber, sensor and laser noise, and sensitivity to environmental conditions such as outside lighting and laser alignment with the sensor. These could be improved by modelling the casing such that the laser and sensor are perfectly aligned and held in place, by shaping the CAD model around the exact shape of the sensor and laser. This variance could also be mitigated in future research by using more complex identification algorithms that look at patterns and changes in readings over time to adjust for lighting or alignment conditions. Future research would also include expanding the scope of the detector to also identify other indoor air pollutants, such as nitrogen monoxide and formaldehyde, which are common in residential settings and often underaddressed.